

Cross-Holdout Aggregation Order and Profile Robustness in a Prime-Shell Diagnostic

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Abstract

We audit a finite common-ambient prime-shell holdout experiment under an explicit family of aggregation rules. The input is the released TPC-309 atlas: three adjacent source-backed profile windows, 18 shell cases per window, and three exclusive-completion radii. Rather than select one profile, we enumerate all 49 nonempty profile-subset/radius-subset selectors and compute 147 aggregate intervals. A pooled mean-square-error ratio, an equal-case arithmetic mean of row ratios, and an equal-case geometric mean give different finite classes. On the full selector the pooled interval is $[0.2423655855, 0.3112477031]$ (right lower), while the arithmetic interval is $[5.2417686281, 14.4871333704]$ (left lower); the geometric interval is $[0.1993188213, 0.8609189559]$ (right lower). We prove the finite interval algebra and the identity that pooled ratios are denominator-weighted means of row ratios. The result is a scoped aggregation-order obstruction, not a causal, asymptotic, arithmetic, or twin-prime theorem.

1 Question and position on the route

The recent finite diagnostics separate increasingly precise sources of instability. TPC-307 introduced a common ambient operator with overlap-only fits and exclusive holdouts [2]. TPC-308 enumerated binary completion envelopes on those holdouts [1]. TPC-309 shifted a source-backed 17-coordinate profile window through a fixed 19-prime pool and found that the location of strict discordance changes with the window [3].

The next minimal question is whether cross-holdout aggregation removes this sensitivity in a way that is independent of a hidden weighting convention. The answer matters before interpreting any finite preference as structural: a pooled loss and an equal-case ratio answer different questions. TPC-310 makes that distinction explicit and audits it exhaustively on the locked parent record.

All numerical inputs below are read from the canonical TPC-309 certificate. No labels, shell partitions, physical source rows, or completion balls are regenerated or changed.

2 Finite protocol

Let i index one of the 162 TPC-309 envelope observations. Its parent certificate supplies a positive ratio interval

$$I_i = [\rho_i^-, \rho_i^+], \quad \rho_i = \frac{\text{right holdout loss}}{\text{left holdout loss}},$$

and positive extrema bounds. Write r_i^- and r_i^+ for the lower bound of the right minimum and the upper bound of the right maximum, respectively; write ℓ_i^- and ℓ_i^+ analogously for the left minimum and maximum.

The profile labels are $\mathcal{L} = \{\text{LOW}, \text{BASE}, \text{HIGH}\}$ and the completion radii are $\mathcal{R} = \{0, 1, 2\}$. We enumerate every pair (L, R) of nonempty subsets of $\mathcal{L}$ and $\mathcal{R}$. There are $7 \times 7 = 49$ selectors. If $S(L, R)$

is the set of parent observations selected by (L, R) , three aggregate maps are declared:

$$P(S) = \left[\frac{\sum_{i \in S} r_i^-}{\sum_{i \in S} \ell_i^+}, \frac{\sum_{i \in S} r_i^+}{\sum_{i \in S} \ell_i^-} \right], \quad (1)$$

$$A(S) = \left[\frac{1}{|S|} \sum_{i \in S} \rho_i^-, \frac{1}{|S|} \sum_{i \in S} \rho_i^+ \right], \quad (2)$$

$$G(S) = \left[\exp\left(\frac{1}{|S|} \sum_{i \in S} \log \rho_i^-\right), \exp\left(\frac{1}{|S|} \sum_{i \in S} \log \rho_i^+\right) \right]. \quad (3)$$

Here P aggregates losses before division, whereas A and G give each parent observation equal weight. An interval is classified as right lower if its upper endpoint is below 0.9, left lower if its lower endpoint is above 1.1, and unresolved otherwise. The budget reference used only for a finite agreement diagnostic is the strict majority of the profile-budget classes in the selected ladders; ties are unresolved.

3 Finite algebra

Lemma 1 (Independent finite extrema). *Let X_i be nonempty finite sets with extrema x_i^- and x_i^+ . If the choices $x_i \in X_i$ are independent, then*

$$\min \sum_i x_i = \sum_i x_i^-, \quad \max \sum_i x_i = \sum_i x_i^+.$$

Proof. Every summand is bounded by its corresponding endpoint, giving the two inequalities. Choosing all lower endpoints, respectively all upper endpoints, attains them. $\square$

Proposition 1 (Soundness of the three maps). *For every nonempty selector S , equations (1)–(3) are positive interval enclosures for their declared finite aggregation rules.*

Proof. The pooled numerator and denominator extrema add by the lemma. Since all denominators are positive, quotient order reverses only in the denominator, giving (1). Arithmetic averaging is coordinatewise monotone. On $(0, \infty)$, both log and exp are increasing, so applying them to the two endpoint sums gives (3). The threshold classification is sound whenever an endpoint lies strictly beyond its declared threshold. $\square$

Proposition 2 (Aggregation-order identity). *For positive point values a_i, b_i and $q_i = a_i/b_i$,*

$$\frac{\sum_i a_i}{\sum_i b_i} = \frac{\sum_i b_i q_i}{\sum_i b_i}.$$

Thus a pooled ratio is a b_i -weighted mean of row ratios, while an equal-case arithmetic ratio is an unweighted mean.

Proof. Substitute $a_i = b_i q_i$ and divide by the positive sum $\sum_i b_i$. $\square$

The identity does not select either map as canonical. It instead predicts that aggregation can reverse a threshold class when row denominators are correlated with row ratios. This is an analytic structure reusable beyond the present finite data.

4 Results

The producer and an independent parser/replay agree on all 49 selectors and 147 aggregate rows. The inherited parent contains 54 profile cases, 162 envelope observations, and 2,106 completion-candidate evaluations.

The full selector uses all three ladders and all three radii. Its intervals are shown in Table 2. The first two are separated from both thresholds by a wide margin, so their reversal is not a decimal-boundary effect.

The profile-singleton slices (all radii) give P = right, right, unresolved for LOW, BASE, HIGH; A = unresolved, left, unresolved; and G = right, right, unresolved. The radius-singleton slices (all profiles)

Table 1: Class census over all 49 selectors (147 aggregate rows).

Aggregation map	Right	Left	Unresolved
Pooled MSE P	42	1	6
Balanced ratio A	1	32	16
Geometric ratio G	26	0	23

Table 2: Full-selector aggregate intervals.

Map	Interval	Class
P	[0.2423655855, 0.3112477031]	Right lower
A	[5.2417686281, 14.4871333704]	Left lower
G	[0.1993188213, 0.8609189559]	Right lower

give P = right at each radius and A = left at each radius, while G is right at radius zero and unresolved at radii one and two. Leave-one-ladder-out pooled slices are all right, but the HIGH-only pooled slice is unresolved and the LOW-only radius-zero slice is right while the HIGH-only radius-zero slice is left.

For a more direct comparison, the pairwise class table over the 49 selectors is: The omitted pair categories have count zero. In particular, P and A are oppositely strict on 29 selectors, and the full selector is one of those reversals.

5 Interpretation and obstruction

The pooled result is dominated by the denominator weights implicit in the loss sum. Using interval midpoints only as a descriptive diagnostic, the full selector has a pooled ratio near 0.5072 but an equal-case ratio near 9.8645; the largest five left-loss weights account for about 38.1% of the total weight, while the largest single radius-zero weight accounts for about 25.7% of that radius slice. These midpoint summaries are not additional theorem claims; they illustrate equation (2).

The finite obstruction is therefore two-layered. First, TPC-309 already showed that changing a profile prefix moves local discordance. Second, cross-holdout aggregation can erase, preserve, or reverse the apparent global orientation depending on whether rows are weighted by loss scale or equally. Consequently, a statement such as “the right completion is preferred” is incomplete until the aggregation measure and its arithmetic justification are fixed before inspecting the data.

6 Claim firewall and conclusion

The exact part of this paper is the selector enumeration, finite extrema rule, positive interval algebra, and weighted-mean identity. The 49-selector atlas is a numerical reproduction of padded parent intervals, not a directed-rounding certificate. Target-generation leakage inherited from TPC-302 remains explicit. We obtain no arithmetic L^2 estimate, no fixed-power credit, no uniform asymptotic budget, no causal identification, no full Gate B passage, and no twin-prime conclusion.

The scoped conclusion is:

On the locked finite TPC-309 atlas, the declared pooled, equal-case arithmetic, and geometric aggregation maps do not share a universal strict class. The full-selector pooled and arithmetic classes reverse. Any future preference claim therefore requires a pre-registered weighting or stratification theorem and an independent holdout replication.

Reproducibility. The canonical certificate is `results/tpc310_certificate.json`. The producer, independent checker, exact rational stress suite, and Bridge-B checker are included with this manuscript. The Session-named Route-A/Route-B evaluator files were absent from the checkout, so no official evaluator pass is asserted.

Table 3: Pairwise class counts; entries are “first | second”.

Pair	R R	R L	R U	L L	L U	U U
$P A$	1	29	12	1	0	4
$P G$	26	0	16	0	1	6
$A G$	1	19	13	0	0	10

References

- [1] Liang Wang. Adversarial exclusive-completion envelopes for a prime-shell holdout, 2026. TPC-308 project release, Huazhong University of Science and Technology.
- [2] Liang Wang. Common-ambient union-shell holdout in a prime-shell diagnostic, 2026. TPC-307 project release, Huazhong University of Science and Technology.
- [3] Liang Wang. Profile-prefix shift sensitivity in a common-ambient prime-shell holdout, 2026. TPC-309 project release, Huazhong University of Science and Technology.